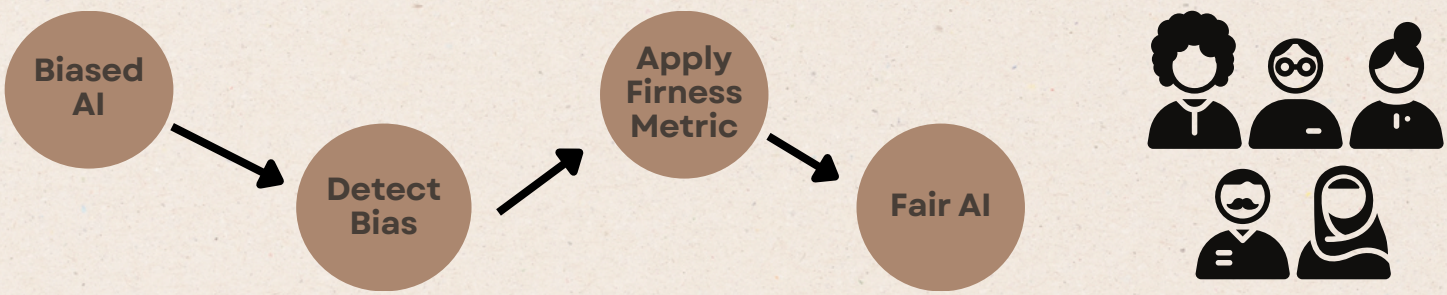
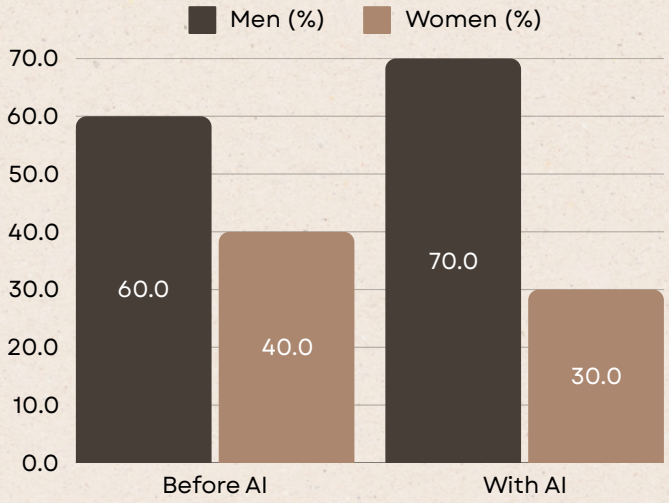
Ensuring Fairness in AI: Group Fairness Metrics

Group fairness in machine learning ensures that AI treats all groups equally, without bias based on gender, race, or age. Since AI learns from past data, it can inherit biases, leading to unfair decisions in hiring, lending, and healthcare. The goal is to detect and correct these biases so AI makes fair and unbiased decisions for all .



Real-World Use Case - Amazon's AI Hiring Bias

In 2018, Amazon's AI hiring tool favored men because it learned from past male-dominated resumes, leading to bias against women. Fairness metrics like Demographic Parity, Equalized Odds, and Calibration can help ensure fair hiring. Due to its bias, Amazon discontinued the tool, highlighting the need for fairness in AI.



Three Group Fairness Metrics

1 - Demographic Parity

Demographic Parity: Ensures equal hiring rates for different groups (e.g., if 50% of men are hired, 50% of women should be too).

- Vs. Equalized Odds: Focuses only on selection rates, not candidate qualifications.
- Impact on Amazon AI: Would have ensured equal hiring but not necessarily the most qualified candidates.

2 - Equalized Odds

Equalized Odds: Ensures AI has the same true positive and false positive rates across all groups, meaning it fairly identifies qualified candidates.

- Vs. Demographic Parity: Considers actual qualifications, while Demographic Parity only looks at selection rates.
- Impact on Amazon AI: Would have reduced ranking bias by evaluating men and women based on their true qualifications.

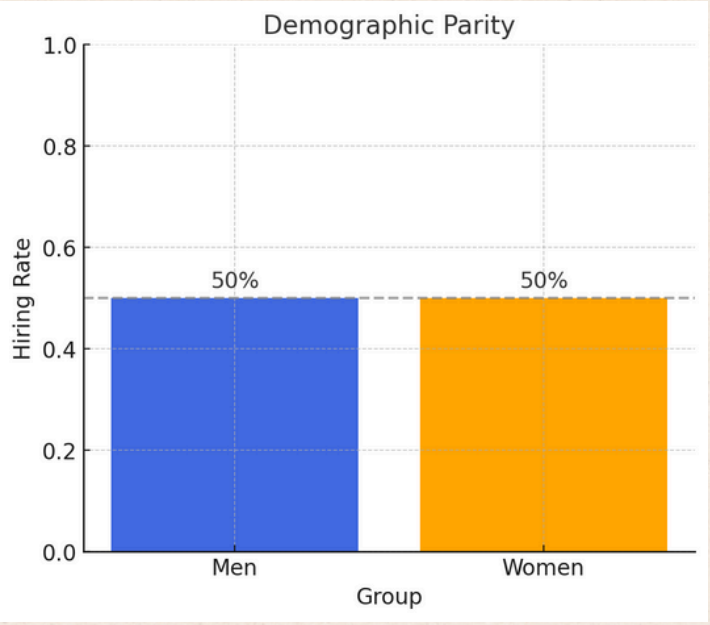
3 - Calibration

Calibration: Ensures AI predictions are equally reliable for all groups. An 80% hiring probability should reflect the same actual success rate for both men and women.

- Vs. Demographic Parity: Focuses on prediction accuracy, not just equal hiring rates.
- Vs. Equalized Odds: Ensures score reliability rather than just balancing error rates.
- Impact on Amazon AI: Would have ensured a woman with an 80% hiring score had the same real success rate as a man, preventing biased rankings.

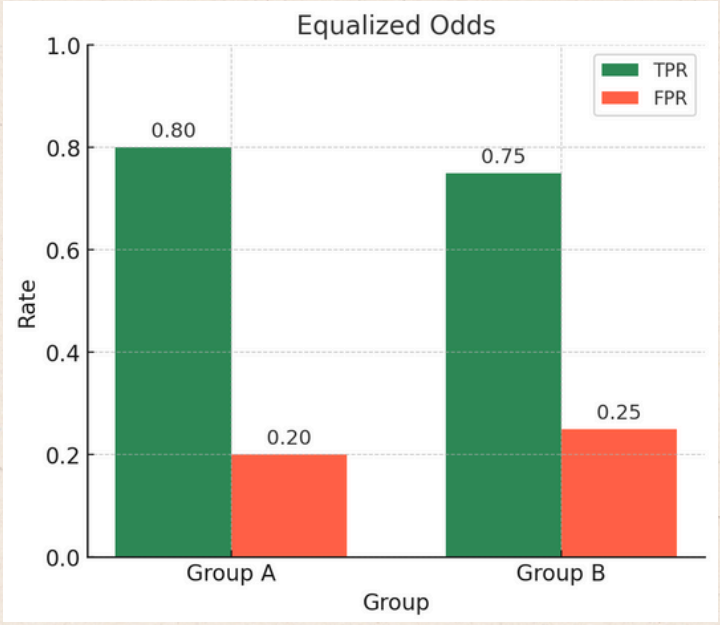
1 - Demographic Parity

- A bar chart showing equal hiring rates (50%) for both groups (Men & Women).
- This ensures equal selection rates regardless of qualifications.



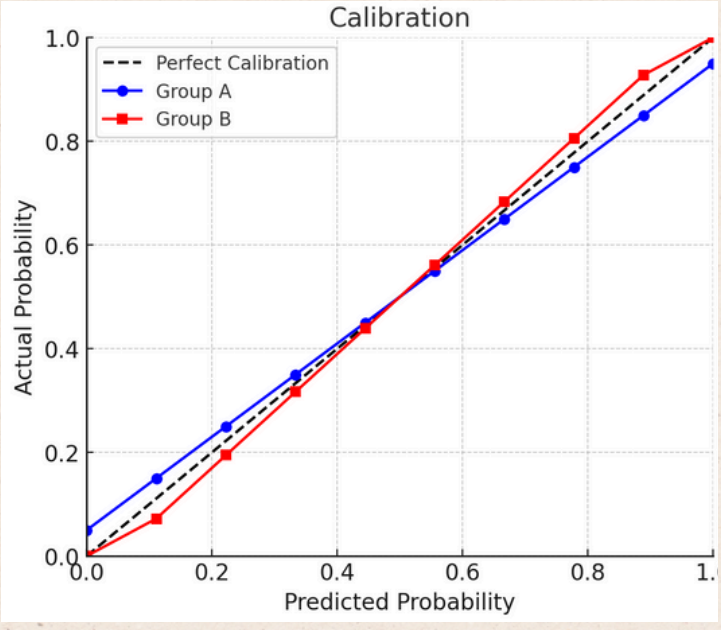
2 - Equalized Odds

- A side-by-side bar chart for True Positive Rate (TPR) and False Positive Rate (FPR) across Group A and Group B.
- If both TPR and FPR match, then Equalized Odds is satisfied.



3 - Calibration

- A calibration curve comparing predicted probability vs. actual probability for two groups.
- The dashed line represents perfect calibration, and the two groups should ideally align with it.



Conclusion

AI should be fair and unbiased to prevent discrimination in critical decisions like hiring and lending. Using fairness metrics such as Demographic Parity, Equalized Odds, and Calibration helps, but no single method is perfect. The best approach is to combine these metrics, continuously monitor AI for bias, and refine it over time. This ensures ethical decision-making and promotes equal opportunities for all.

References

AXA's The Fairness Compass (Fairness Metrics):

- The Fairness Compass: A Groundbreaking Step Forward for Trustworthy AI
- Towards the Right Kind of Fairness in AI (PDF)

Amazon AI Hiring Bias Case Study (2018):

- Amazon scraps secret AI recruiting tool that showed bias against women
- Amazon Built AI to Hire People, but It Discriminated Against Women

Ethical AI Research Papers:

- Exploring Bias and Fairness in Artificial Intelligence and Machine Learning Algorithms
- Fairness and Bias in Artificial Intelligence: A Brief Survey of Sources, Impacts, and Mitigation Strategies

Tools Used:

- Canva